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ABOUT

Physics graduate specializing in quantum systems and machine learning, with published research in Physical Review C and Machine Learning: Science and Technology.

- INTERESTS
- Quantum Many-Body Physics
 - Machine Learning
 - Computational Physics
 - Quantum Computing
 - Condensed Matter Physics

01 Academic Background

- M.Sc. in Physics, The University of Iowa

Aug. 2024 -- Aug. 2025

GPA 4.07/4.33. Supervisor: Yannick Meurice. Funding: teaching and research assistantships. Thesis: *Predicting the von Neumann Entanglement Entropy Using a Graph Neural Network*.
- B.Sc. in Physics, The University of Jordan

Sept. 2019 -- Jun. 2023

GPA 3.93/4.00 (rank 2 of 42). Graduation project: *A Phenomenological Study of the α -Nucleus Elastic Scattering Using a Velocity-Dependent Optical Model*.

02 Publications

- P1

Predicting the von Neumann Entanglement Entropy Using a Graph Neural Network

Machine Learning: Science and Technology, 2025 · Sole author

Master’s thesis work. [arXiv: 2503.23635](#) · [DOI: 10.1088/2632-2153/adf811](#) · [code](#)
- P2

An Interpretable Closed Form for Entanglement Entropy from Bitstrings, Guided by a Graph Neural Network

Preprint, 2026 · Sole author

[arXiv: 2606.22713](#)
- P3

Zeros of the Partition Function for 12-Flavor QCD

Preprint, 2026 · First author · with M. Hite, D. Floor, Y. Meurice

[arXiv: 2606.14642](#)
- P4

Modeling α -Nucleus Elastic Scattering Using a Velocity-Dependent Optical Potential

Physical Review C, 2024 · First author

Undergraduate thesis. [arXiv: 2310.11816](#) · [DOI: 10.1103/PhysRevC.109.034606](#)

03 Presentations

Poster: “Predicting von Neumann Entanglement Entropy with Graph Neural Networks,” 11th International Petra School of Physics, Amman, Jordan, October 2025.

04 Research Experience

- Master’s Thesis

Sept. 2024 -- Aug. 2025

Developed a graph neural network to predict von Neumann entanglement entropy in Rydberg atoms and Ising models, outperforming mutual information estimation and generalizing to larger systems via fine-tuning.
- Interpretable Closed Form for Entanglement Entropy

Apr. 2026 -- Present

Distilled a trained graph neural network into a compact six-feature closed-form estimator of the von Neumann entanglement entropy, validated against measurement noise, distribution shifts, and larger system sizes.
- Fisher Zeros Analysis in Lattice Gauge Theories

May. 2025 -- Present

Studying the scaling of Fisher zeros in the partition function of multi-flavor lattice QCD.
- Quantum Many-Body Scars in the Compact Abelian Higgs Model

Jul. 2025 -- Feb. 2026

Studied the non-thermalization of specific initial states in the compact Abelian Higgs model.
- Final Year Undergraduate Research Project

Jul. 2022 -- May. 2023

Modeled α -nucleus elastic scattering using a velocity-dependent optical potential, significantly improving predictions of anomalous large-angle scattering (ALAS).
- Research Assistant, Magnetic Nanoparticles Laboratory

Aug. 2022 -- Sept. 2022

University of Jordan. Engaged in sample preparation and analysis.

05 Employment History

- AI Engineer, Logatta

Building and deploying commercial AI tools at a technology company specializing in the implementation and development of AI products.

Apr. 2026 -- Present
- Research Assistant, University of Iowa

Analyzed Fisher zeros in multi-flavor lattice QCD.

Jun. 2025 -- Jul. 2025
- Teaching Assistant, University of Iowa

Served as a teaching assistant for various laboratories in the physics department.

Aug. 2024 -- May. 2025
- Part-Time Lecturer & TA, University of Jordan

Taught a senior-level Python course for physics undergraduates; served as a teaching assistant for various laboratories.

Jun. 2023 -- Jun. 2024

06 Technical Skills

Languages	Python, C++, Mathematica
Machine learning	PyTorch, PyTorch Geometric
Compute	CUDA; HPC / Slurm clusters
Nuclear scattering	FRESCO
Molecular dynamics	Simulations using Amber

07 Administrative Committee Involvement

- Graduate Representative

Higher Committee for International Accreditation for the College of Science, University of Jordan. Participated in accreditation processes and ensured compliance with international standards.
- Member

Student Communication Committee, University of Jordan. Facilitated communication between students and faculty, addressing student concerns and advocating for student feedback.

08 Extracurricular Activities

- School Organizing Committee, Scientists for Palestine

Helped organize the Palestinian Advanced School on Computational Methods for Displaced Students from Gaza (2026) and the Palestinian Advanced Physics School (2025).
- Coordinator, 10th International Petra School of Physics

Assisted in organizing this quantum-computing-focused school held in Jordan in 2023.
- Co-Founder & Head of Organizational Committee, Hilbert Science Club

University of Jordan. Established a community for physics and mathematics students to explore emerging topics in their fields.